TOWARD THE TWIN PRIME CONJECTURE: CERTIFICATES, LIMITS OF THE SIEVE, AND A WORKING PROGRAMME

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ABSTRACT. This is a living working paper. It does *not* prove the twin prime conjecture, and it does not improve the best known bound on small gaps between primes. It records (i) the state of the problem as of the date above, (ii) what this project has verified for itself, by exact computation and by the Lean 4 kernel, now including the established bound $H_1 \le 246$ of Polymath8b, reproduced end to end through a certified lower bound for $M_{50,1/25}$, (iii) small results of our own: a no-go statement for a natural attempt to sharpen the pigeonhole step, on the plain and on the ε-enlarged simplex; a generating-function formula that makes certified computation with all symmetric test functions cheap; an upper bound showing that the ε-enlargement of Polymath8b cannot, by itself, reach any $k \le 29$; numerical evidence that this upper bound is close to the limit of every dual weight depending only on t_m and $t_1+\cdots+t_k$, and heuristic evidence that the ε-enlarged simplex stops at $k = 50$, and (iv) a programme of original directions that later versions will test. Every statement carries one of the labels LEAN-CHECKED, COMPUTED, PROVED HERE, LITERATURE, CLAIMED (a preprint not yet independently verified), or OPEN.

1. The problem and the honest position

Write p_n for the n-th prime and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. The twin prime conjecture is the statement $H_1 = 2$. It is OPEN.

A finite set $\mathcal{H} = \{h_1 < \cdots < h_k\}$ of integers is *admissible* if for every prime p it misses at least one residue class modulo p. Let DHL$[k, 2]$ be the statement that every admissible k-tuple has infinitely many translates $n + \mathcal{H}$ containing at least two primes, and let $H(k)$ be the least diameter $h_k - h_1$ of an admissible k-tuple. Then

$$\text{DHL}[k, 2] \implies H_1 \le H(k), \tag{1}$$

and DHL$[2, 2]$ applied to $\{0, 2\}$ is the twin prime conjecture. All progress on H_1 since 2013 consists of proving DHL$[k, 2]$ for smaller k.

Where the bound stands (12 November 2026).

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$H_1 \leq$	k	Source	Label
7×10^7	3.5×10^6	Zhang [20]	LITERATURE
600	105	Maynard [10]	LITERATURE
246	50	Polymath8b [14]	LITERATURE
240	49*	Stadlmann [18]	CLAIMED
236	48*	Kintali [8]; Song–Yue [17]	CLAIMED
212	45	Charton et al. [3]	CLAIMED
186	40	OpenAI [13]	CLAIMED
182	39	GPT-6 Astra, prompted by Sharifi [15]	CLAIMED
12	5	Maynard, assuming EH [10]	LITERATURE
6	3	Polymath8b, assuming GEH [14]	LITERATURE

* Value of k inferred by us from $H(49) = 240$ and $H(48) = 236$, not read from the source.

The bound 246 stood from 2014 until the last days of August 2026. It has been formalised in Lean 4, assuming only the Bombieri–Vinogradov theorem [1]. The bounds from 240 to 186 appeared within one week. They rest on Stadlmann's idea of enlarging the support of the sieve weights using equidistribution estimates for moduli with a large smooth factor. The Lean development accompanying the bound 186 is conditional on three stated inputs (two exponential-sum bounds and a family of numerical integral bounds) [13]. An independent re-run of that repository, dated 7 and 8 November 2026, reproduced the build and the numerical certificate and describes the result as "conditionally supported, with the computation independently reproduced; not yet settled"; the two exponential-sum bounds remain axioms [12]. A further document, dated 18 September 2026 and naming an AI system as its author, claims DHL$[39, 2]$ and hence 182; by its own abstract its Lean development verifies the last implication "under explicit numerical and finite-field premises" [15]. We had missed it in earlier versions. (These details were read by an earlier run of this project; on 12 November the file was too large for our reader and direct download was refused, so we have not re-checked them since.) As of 7 September 2026 a standard reference still listed 246 as the best fully established value [11], and the page read the same on 12 November. On the date of this version we found no report of complete independent verification of any of the 2026 claims and no claimed bound below 182. We therefore keep two columns throughout: *established* (246) and *claimed* (182).

Why 2 is a different problem. Under the generalised Elliott–Halberstam conjecture the sieve gives $H_1 \leq 6$, and Polymath8b showed that 6 cannot be improved by purely sieve-theoretic arguments: this is Selberg's parity obstruction [14]. Lowering the bound (Track A below) and proving the conjecture (Track B) are therefore different tasks. No amount of success on Track A reaches 2.

2. What this project has verified for itself

2.1. **The combinatorial half, checked by the Lean kernel.** Our Lean development is written in core Lean 4, without Mathlib. It defines primality, the twin prime conjecture, the predicate `BoundedGaps B` (infinitely many pairs of distinct primes at distance at most B), admissibility, and DHL$[k, 2]$. It contains no `sorry`, and the only axioms it uses are `propext` and `Quot.sound`. The statement DHL$[k, 2]$ is always a hypothesis, never an axiom.

Theorem 1 (LEAN-CHECKED)**.** (1) *The twin prime conjecture is equivalent to* `BoundedGaps 2`.

(2) *If a finite Boolean check succeeds on a list $\mathcal{H}$ (it misses a class modulo every m with $2 \le m \le |\mathcal{H}|$), then $\mathcal{H}$ is admissible. The moduli $p > |\mathcal{H}|$ are handled by a formal pigeonhole argument.*

(3) *For each pair (k, D) in Table 1, for $(k, D) = (56, 278)$, $(57, 282)$, $(58, 288)$, $(59, 300)$, $(60, 304)$, $(61, 310)$, $(62, 320)$, $(63, 324)$, $(64, 330)$, and for $(102, 578)$ and $(105, 600)$, there is an explicit admissible k-tuple in $[0, D]$, and* DHL$[k, 2]$ *implies* `BoundedGaps D`*. In particular* DHL$[2, 2]$ *implies the twin prime conjecture.*

(4) DHL$[k, 2]$ *implies* DHL$[k', 2]$ *for every $k' \ge k$ (`DHL.mono`: delete shifts from the tuple). So a proof of* DHL$[j, 2]$ *with $j \le k$ gives* `BoundedGaps D` *for each listed pair (k, D).*

(5) `BoundedGaps 1` *is false (`not_boundedGaps_one`), and a pair $\{0, d\}$ is admissible if and only if d is even (`admissible_pair_iff`). So 2 is the smallest bound of this form that can hold, and the twin prime conjecture is the strongest statement of the form* `BoundedGaps D`*.*

TABLE 1. Pairs (k, D) with a kernel-checked admissible k-tuple of diameter D. The tuples were found by our own search; each D agrees with the value of $H(k)$ in published tables, whose optimality we have not re-proved.

k	D	k	D	k	D	k	D	k	D	k	D
2	2	11	36	20	80	29	130	38	176	47	226
3	6	12	42	21	84	30	136	39	182	48	236
4	8	13	48	22	90	31	140	40	186	49	240
5	12	14	50	23	94	32	146	41	188	50	246
6	16	15	56	24	100	33	152	42	196	51	252
7	20	16	60	25	110	34	156	43	200	52	254
8	26	17	66	26	114	35	158	44	210	53	264
9	30	18	70	27	120	36	162	45	212	54	270
10	32	19	76	28	126	37	168	46	216	55	272

Theorem 1 is elementary. Its value is that implication (1) is now a checked interface: any future proof of DHL$[k, 2]$, by anyone, yields the corresponding gap bound with nothing left to trust on the combinatorial side.

2.2. **The sieve constant, with exact certificates.** Let $\mathcal{R}_k = \{t \in [0,1]^k : t_1 + \cdots + t_k \le 1\}$ and, for square-integrable F supported on $\mathcal{R}_k$,

$$I_k(F) = \int_{\mathcal{R}_k} F^2, \qquad J_k^{(m)}(F) = \int \left(\int F \, dt_m \right)^2 \prod_{i \ne m} dt_i, \qquad M_k = \sup_F \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

Maynard's theorem [10, Prop. 4.2] states that if the primes have level of distribution θ and $\theta M_k/2 > 1$ then DHL$[k, 2]$ holds. With Bombieri–Vinogradov ($\theta < 1/2$) the condition is $M_k > 4$.

We take $F = \sum_{a+2b \le d} c_{a,b} (1 - P_1)^a P_2^b$ with $P_j = \sum_i t_i^j$, as in [10, §8]. All integrals reduce to

$$\int_{\mathcal{R}_k} (1 - P_1)^A P_2^B \, dt = \frac{A!}{(k + A + 2B)!} \, G_k(B), \qquad \sum_{B \ge 0} G_k(B) \frac{x^B}{B!} = \left(\sum_{b \ge 0} \frac{(2b)!}{b!} x^b \right)^k,$$

which follows from the Dirichlet integral. The two quadratic forms are then exact rational matrices. We compute an approximate top eigenvector in floating point, round it to rationals,

and evaluate the Rayleigh quotient in exact arithmetic; the last step is a rigorous lower bound whatever happened before it.

Theorem 2 (Computed, exact rational arithmetic)**.** *With F a polynomial in $1-P_1$ and P_2 only,*

$$M_{105} \geq 4.00206976, \qquad M_{102} \geq 4.00271343, \qquad M_5 \geq 2.00410541.$$

The first value reproduces Maynard's $M_{105} > 4$ with $d = 11$ (42 coefficients). The third gives DHL$[5,2]$ and $H_1 \leq 12$ under the Elliott–Halberstam conjecture, again reproducing [10]. For $d = 11, 15, 19$ the least k at which this family certifies $M_k > 4$ is $105, 102, 102$, and at $k = 101$ it gives only 3.99931 ($d = 19$): polynomials in P_1, P_2 alone stop at $k = 102$.

2.3. **All symmetric test functions.** To go further we allow every symmetric polynomial. Since $P_1, \dots, P_k$ generate the symmetric polynomials, the functions

$$(1-P_1)^a P^b := (1-P_1)^a \prod_{j \geq 2} P_j^{b_j}, \qquad a + w(b) \leq d, \quad w(b) = \sum_j j b_j,$$

span all symmetric polynomials of degree at most d (for $k \geq d$). The integrals have a closed form that we have not seen stated in this way (Proved here; possibly folklore).

Lemma 3. *Let $X_1, \dots, X_k$ be independent exponential random variables of mean 1 and $E_k(b) = \mathbb{E} \prod_j P_j(X)^{b_j}$. Then*

$$\int_{\mathcal{R}_k} (1-P_1)^A P^b \, dt = \frac{A!}{(k+A+w(b))!} E_k(b), \qquad \sum_b E_k(b) \frac{s^b}{b!} = \left(\sum_c w(c)! \, \frac{s^c}{c!} \right)^k,$$

where $s^b = \prod_j s_j^{b_j}$ and $b! = \prod_j b_j!$. In particular each $E_k(b)$ is a positive integer, computed exactly by multiplying exponential generating functions.

Proof. For a monomial, the Dirichlet integral gives $\int_{\mathcal{R}_k} (1-P_1)^A t^\alpha \, dt = A!\,\alpha!/(k+A+|\alpha|)!$, and $\alpha! = \mathbb{E} X^\alpha$. Summing over the monomials of P^b, all of degree $w(b)$, gives the first identity. For the second, $\mathbb{E} \exp(\sum_j s_j P_j(X)) = \left(\mathbb{E} \exp(\sum_j s_j X_1^j) \right)^k$ by independence, and $\mathbb{E} \exp(\sum_j s_j X_1^j) = \sum_c \mathbb{E} X_1^{w(c)} \, s^c/c! = \sum_c w(c)! \, s^c/c!$, as formal power series. □

The J-functional reduces to the same lemma in $k-1$ variables: with $P_j = P_j' + t_k^j$,

$$\int_0^{1-P_1'} (1-P_1)^a P^b \, dt_k = \sum_{e \leq b} \binom{b}{e} \frac{a! \, w(e)!}{(a+w(e)+1)!} (1-P_1')^{a+w(e)+1} P'^{\,b-e}.$$

So both Gram matrices are exact rational matrices. We find an approximate top eigenvector by inverse iteration with a fixed shift above $\frac{k}{k-1} \log k$, which can only converge to the top eigenvalue. We take its floating-point midpoints as exact dyadic rationals and evaluate the Rayleigh quotient exactly. The working precision of the eigenvector matters only for speed. A separate script, `code/verify_cert.py`, rebuilds both Gram matrices from scratch and re-evaluates each stored coefficient vector in exact arithmetic.

Theorem 4 (Computed, exact rational arithmetic)**.**

$$M_{105} \geq 4.17427, \qquad M_{64} \geq 4.00363, \qquad M_{58} \geq 4.00927,$$
$$M_{56} \geq 4.00519, \qquad M_{55} \geq 4.00516, \qquad M_{54} \geq 4.00238.$$

The first two use all symmetric polynomials of degree 11 and 12 (195 and 272 coefficients). The others use the subspace spanned by $(1-P_1)^a P^b$ with b supported on even j, as in [14, §7]*, of degree 16, 18, 20 and 23 (307, 471, 707 and 1236 coefficients).*

Basis	degree	coefficients	least k with certified $M_k > 4$
$1 - P_1$, P_2	19	110	102
all symmetric	12	272	64
even power sums	16	307	58
even power sums	18	471	56
even power sums	20	707	55†
even power sums	23	1236	54†

† Only this value of k was run at this degree; smaller k were not tried.

Corollary 5 (end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [10, Prop. 4.2]*, both* LITERATURE*, Theorems 4 and 1 give* $H_1 \le 270$*.*

This reproduces a known result and is not new. Polymath8b obtained $M_{54} > 4.00238$ with even-part polynomials of degree 23 [14, Thm. 3.9], and hence $H_1 \le 270$. Our certified value in the same space, $4.002386\ldots$, agrees with theirs to every digit they print, which is an external check on our pipeline; two independent runs of it (one from an earlier session) produced the same value. Corollary 5 is the best that the plain simplex gives with Bombieri–Vinogradov (see the next subsection); Corollary 7 below takes the next rung of the ladder $600 \to 270 \to 246 \to 240 \to 212 \to 186 \to 182$.

2.4. **What the plain method cannot do.** Polymath8b proved $M_k \le \frac{k}{k-1}\log k$ [14]. Hence (COMPUTED from that bound) $M_k > 4$ forces $k \ge 51$, and on the plain simplex DHL$[k,2]$ requires a level of distribution $\theta > 2(k-1)/(k\log k)$:

k	50	45	40	35	30	20	10	5
$\theta >$	0.501	0.514	0.529	0.547	0.568	0.634	0.782	0.994

The 2026 results go below $k = 50$ by changing the support of F, not by proving a uniform level of distribution of this size.

2.5. **Beyond the plain simplex:** $H_1 \le 246$**.** Polymath8b's ε-enlarged simplex [14, Thm. 3.12] (LITERATURE; we re-read the statement in the arXiv text for this version) takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and integrates the square of each marginal $\int F\,dt_m$ only over $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. If EH$[\theta]$ holds with $1+\varepsilon < 1/\theta$ and $M_{k,\varepsilon} > 2/\theta$, then DHL$[k,2]$ holds. Bombieri–Vinogradov gives every $\theta < 1/2$, so for every fixed $\varepsilon < 1$ the condition is $M_{k,\varepsilon} > 4$.

We extended our pipeline to this functional, following the route that [14, §7.2] describes in words. Put $L = 1+\varepsilon$, $\tau = 1-\varepsilon$ and use the basis $(L-P_1)^a P^b$ (b supported on even j). The following identities are elementary (PROVED HERE): the substitution $t = Ls$ and Lemma 3 give

$$\int_{L\mathcal{R}_k} (L-P_1)^A P^b\,dt = L^{k+A+w(b)}\,\frac{A!}{(k+A+w(b))!}\,E_k(b);$$

the inner integral over $t_m \in [0, L-P_1']$ is the formula displayed after Lemma 3 with $1-P_1'$ replaced by $L-P_1'$; and on $\tau\mathcal{R}_{k-1}$ we write $L-P_1' = 2\varepsilon + (\tau - P_1')$, so that

$$\int_{\tau\mathcal{R}_{k-1}} (L-P_1')^A P'^B = \sum_{i=0}^{A}\binom{A}{i}(2\varepsilon)^{A-i}\,\tau^{k-1+i+w(B)}\,\frac{i!}{(k-1+i+w(B))!}\,E_{k-1}(B).$$

For rational ε both Gram matrices are again exact rational matrices. We checked the code against hand computation for $k = 2, 3$ and $F \in \{1, L-P_1\}$, and against direct iterated symbolic integration (sympy, without Lemma 3 or the binomial rewrite) of every Gram entry for $(k, d, \varepsilon) = (3, 3, 1/5)$ and $(4, 4, 1/25)$: no mismatch (`code/check_eps_sympy.py`). At $\varepsilon = 0$ it is the old

code path (an old certificate re-verifies unchanged). We also found that the working precision of the eigenvector search can be cut from about 6000 to 300–400 bits without changing any certified digit, which makes degree 25 feasible on a four-core machine (build 16 minutes, search 4 minutes).

Theorem 6 (Computed, exact rational arithmetic)**.** *With even power sums of degree* 25 *(*1780 *coefficients) and of degree* 27 *(*2526 *coefficients),*

$$M_{50,1/25} \geq 4.0012466 \qquad \textit{and} \qquad M_{50,1/25} \geq 4.0043063.$$

The certificates are `code/cert_M50e25_D25e.json` and `code/cert_M50e25_D27e.json` (the degree-27 run took 43 minutes to build and 10 minutes to search, with about 10 GB of memory); the script `code/verify_cert.py` rebuilds both Gram matrices from scratch and re-evaluates the stored rational vector exactly; it uses the same Gram-matrix builder, so it checks the arithmetic and the bookkeeping, while the formulas themselves are checked by the symbolic test above. Polymath8b report $M_{50,1/25} > 4.00124$ at degree 25 and > 4.0043 at degree 27 [14, §7.2]; our values agree with both to every printed digit.

Corollary 7 (end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [14, Thm. 3.12(i)]*, both* Literature*, Theorem 6 and the Lean tuple* $(50, 246)$ *of Theorem 1 give* $H_1 \leq 246$.

This is the established bound of 2014, reproduced; it is not new. It closes the second rung of our ladder. Table 2 shows how the certified values grow with the degree and depend on ε; it is the data behind Remark 20 below.

Table 2. Certified lower bounds for $M_{k,\varepsilon}$ (Computed, exact Rayleigh quotients, even power sums of degree D). Data: `runs/scan_k49.jsonl`, `runs/scan_k50.jsonl`, written by `code/scan_eps.py`; the column $k = 50$, $D = 16$ from direct runs of `code/mk_general.py`. A dash: not computed.

	$k = 49$			$k = 50$		
ε	$D = 16$	$D = 18$	$D = 20$	$D = 16$	$D = 18$	$D = 20$
0	3.88640	3.90102	3.91056	3.90173	3.91718	3.92739
1/50	3.93491	3.95066	3.96110	3.94979	3.96644	3.97761
1/30	3.93969	3.95745	3.96933	–	3.97254	3.98522
1/25	3.93674	3.95563	3.96840	3.95041	3.97034	3.98395
1/20	3.92741	3.94823	3.96244	–	3.96234	3.97745

3. Results of our own: rigidity of the pigeonhole step

Every bound above ends with the same step: if the weighted average number of primes among $n + h_1, \ldots, n + h_k$ exceeds 1, two of them are prime, and they are at most $h_k - h_1$ apart. A natural hope is to do better with unequal weights. Fix D, and give each position a weight $c_i \geq 0$. Call a set of positions *D-separated* if any two of its shifts differ by more than D, and let $\alpha_D(c)$ be the largest total weight of a D-separated set. If no two primes among the $n + h_i$ are within D, then $\sum_i c_i \mathbf{1}_{n+h_i \text{ prime}} \leq \alpha_D(c)$. Writing

$$M_k[c] = \sup_F \frac{\sum_m c_m J_k^{(m)}(F)}{I_k(F)},$$

the sieve therefore yields $H_1 \le D$ as soon as $\frac{\theta}{2} M_k[c] > \alpha_D(c)$. For $c = (1, \dots, 1)$ and $D = h_k - h_1$ this is the usual criterion. Could a cleverer c certify some $D < h_k - h_1$ that no sub-tuple certifies?

Proposition 8 (Proved here; possibly folklore)**.** *Let ω be the largest number of shifts h_i contained in an interval of length D. Then for every $c \ge 0$, not identically zero,*

$$\frac{M_k[c]}{\alpha_D(c)} \le M_\omega.$$

So the weighted criterion never certifies more than the standard criterion applied to the best sub-tuple of diameter at most D.

Proof. Let G be the graph on $\{1, \dots, k\}$ joining $i \ne j$ when $|h_i - h_j| \le D$. Its stable sets are the D-separated sets and its cliques are the sets of shifts lying in an interval of length D. Normalise $\alpha_D(c) = 1$, so c lies in

$$P = \{c \ge 0 : \ c(S) \le 1 \text{ for every stable set } S\}.$$

The graph G is a unit interval graph, hence perfect, and so is its complement by the perfect graph theorem [9]. For a perfect graph the polytope cut out by the clique inequalities is the convex hull of the indicator vectors of its stable sets [5]. Applying this to the complement of G gives

$$P = \operatorname{conv}\{\mathbf{1}_K : \ K \text{ a clique of } G\}.$$

The map $c \mapsto M_k[c]$ is a supremum of linear functionals, hence convex, and therefore $M_k[c] \le \max_K M_k[\mathbf{1}_K]$. It remains to show $M_k[\mathbf{1}_K] \le M_j$ for $|K| = j$; this suffices because M_j is non-decreasing in j (adjoin a variable and concentrate F near its zero set). Write $t = (s, u)$ with s the coordinates in K and u the others, put $\sigma(u) = 1 - \sum u \ge 0$, and let $F_u = F(\cdot, u)$, a function on the scaled simplex $\sigma(u)\mathcal{R}_j$. For $m \in K$ we have $J_k^{(m)}(F) = \int J_j^{(m)}(F_u)\, du$, and $I_k(F) = \int I_j(F_u)\, du$. Rescaling F_u to $\mathcal{R}_j$ multiplies I_j by σ^j and each $J_j^{(m)}$ by σ^{j+1}, so

$$\sum_{m \in K} J_j^{(m)}(F_u) \ \le \ \sigma(u)\, M_j\, I_j(F_u) \ \le \ M_j\, I_j(F_u).$$

Integrating over u gives the claim. □

Remark 9 (Computed, consistency check for $k = 3$). With unrestricted polynomials of degree 6 on $\mathcal{R}_3$ we find the lower bounds $M_3[(1,1,1)] \ge 1.6464$, $M_3[(1,1,0)] \ge 1.3579$ and $M_3[(\frac{1}{2}, 1, \frac{1}{2})] \ge 1.1604$, against $M_2 = 1/(1 - W(1/e)) = 1.3859\ldots$. The half-weighted functional is far below M_2, as the proposition requires.

The same rigidity on the enlarged simplex. The ε-enlarged simplex of Polymath8b [14, Thm. 3.12] takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and restricts the outer integral of each $J^{(m)}$ to $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. The criterion $M_{k,\varepsilon} > 2/\theta$ is valid in two cases (Literature): under EH[θ] when $1 + \varepsilon < 1/\theta$, and under GEH[$\theta$] when $\varepsilon < 1/(k-1)$. With Bombieri–Vinogradov the first case allows every $\varepsilon < 1$, and Polymath8b's $H_1 \le 246$ uses $M_{50,1/25} > 4.0043$ [14, Thm. 3.13]. Unequal weights are no more useful there.

Proposition 10 (Proved here; possibly folklore)**.** *For $0 \le \varepsilon < 1$ and every weight $c \ge 0$, $c \ne 0$, $M_{k,\varepsilon}[c]/\alpha_D(c) \le M_{\omega,\varepsilon}$, with equality for the indicator of a largest clique. Since the conditions on ε above do not get harder as k decreases, the weighted criterion never certifies more than the standard one for the best sub-tuple of diameter at most D.*

Proof. As for Proposition 8, it suffices to show $\sum_{m \in K} J_{k,\varepsilon}^{(m)}(F) \le M_{j,\varepsilon} I_k(F)$ for a clique K of size j, together with $M_{j,\varepsilon} \le M_{\omega,\varepsilon}$. Fix the coordinates u outside K, with $|u| \le 1 - \varepsilon$ (other slices do not meet the J-region). The slice F_u is supported on $(1 + \varepsilon - |u|)\mathcal{R}_j$, and its J-region

is $(1-\varepsilon-|u|)\mathcal{R}_{j-1}$. Put $\mu=(1+\varepsilon-|u|)/(1+\varepsilon)\le 1$ and $H(x)=F_u(\mu x)$, which is supported on $(1+\varepsilon)\mathcal{R}_j$. Then $I_j(F_u)=\mu^j I_j(H)$, and each slice J-term equals μ^{j+1} times the integral of $(\int H\,dx_m)^2$ over $|x'|\le(1-\varepsilon-|u|)/\mu$. Since

$$\frac{(1-\varepsilon-|u|)(1+\varepsilon)}{1+\varepsilon-|u|}\le 1-\varepsilon \iff -|u|(1+\varepsilon)\le -|u|(1-\varepsilon),$$

that region lies inside $(1-\varepsilon)\mathcal{R}_{j-1}$, so the slice contributes at most $\mu^{j+1}M_{j,\varepsilon}I_j(H)=\mu M_{j,\varepsilon}I_j(F_u)$. For $j=1$ use Cauchy–Schwarz, which gives $M_{1,\varepsilon}=1+\varepsilon$. Integrate over u and use $\mu\le 1$. Monotonicity in j follows by embedding: take $F(s,u)=H(s/\mu_\delta)\mathbf{1}_{[0,\delta]^{k-j}}(u)$ with $\mu_\delta=1-(k-j)\delta/(1+\varepsilon)$ and let $\delta\to 0$. □

Slicing naively gives each slice its own parameter $\varepsilon'=\varepsilon/(1-|u|)\ge\varepsilon$, which would only bound the weighted constant by $\sup_{\varepsilon'\ge\varepsilon}M_{j,\varepsilon'}$, a weaker statement. The single rescaling by μ above avoids that loss.

Remark 11 (Computed, $k=3$). With unrestricted polynomials (degree 5 for $k=3$, degree 10 for $k=2$, cut off at $|t|\le\rho$ for $\rho\in\{1-\varepsilon,1,1+\varepsilon\}$):

ε	$M_{2,\varepsilon}\ge$	$M_{3,\varepsilon}[(1,1,0)]\ge$	$M_{3,\varepsilon}[(\frac12,1,\frac12)]\ge$	$M_{3,\varepsilon}\ge$
0	1.385933	1.349844	1.160121	1.646427
1/10	1.518567	1.482143	1.264698	1.793492
1/5	1.627161	1.585674	1.341926	1.882282
1/3	1.717104	1.663743	1.413309	1.917001

Both weighted columns stay below $M_{2,\varepsilon}$, as the proposition requires.

Remark 12 (scope). Neither proposition covers more general polytope supports [14, §3], Stadlmann's hybrid supports [18], J-regions that depend on m, or arguments that are not linear in the indicators of primality. Those are Open.

4. Results of our own: how far the ε-enlargement alone can go

For the plain simplex the bound $M_k\le\frac{k}{k-1}\log k$ shows where the method stops ($k\ge 51$). This section starts the same analysis for the ε-enlarged simplex. For $0<\tau\le L$ put

$$M_k(L,\tau)=\sup_F\frac{\sum_{m=1}^k\int_{\tau\mathcal{R}_{k-1}}\left(\int_0^\infty F\,dt_m\right)^2\prod_{i\ne m}dt_i}{\int F^2},$$

the supremum over square-integrable $F\not\equiv 0$ supported on $L\mathcal{R}_k$. Then $M_{k,\varepsilon}=M_k(1+\varepsilon,1-\varepsilon)$ and $M_k=M_k(1,1)$. The quantity $M_k(L,\tau)$ is non-decreasing in L and in τ, satisfies $M_k(cL,c\tau)=c\,M_k(L,\tau)$, and is non-decreasing in k by the embedding used in the proof of Proposition 10. Write $S=t_1+\cdots+t_k$ and call the index m *active* at t if $S-t_m\le\tau$.

Lemma 13 (Proved here; it is [14, Lemma 6.1] with the region of the marginals kept)**.** *Let $\psi_1,\dots,\psi_k$ be measurable functions on $L\mathcal{R}_k$, positive almost everywhere, such that for almost every $t'\in\tau\mathcal{R}_{k-1}$ and every m*

$$\int_0^{L-|t'|}\frac{du}{\psi_m(t',u)}\le 1,$$

where (t',u) is the point whose m-th coordinate is u and whose other coordinates are t'. Then

$$M_k(L,\tau)\le\operatorname*{ess\,sup}_{t\in L\mathcal{R}_k}\sum_{m\ \text{active at}\ t}\psi_m(t).$$

Proof. For $t' \in \tau\mathcal{R}_{k-1}$ the Cauchy–Schwarz inequality gives

$$\left(\int F\,dt_m\right)^2 \le \int \frac{dt_m}{\psi_m} \cdot \int \psi_m F^2\,dt_m \le \int \psi_m F^2\,dt_m.$$

Integrating over $t' \in \tau\mathcal{R}_{k-1}$ gives the integral of $\psi_m F^2$ over the set where m is active. Sum over m. □

Proposition 14 (Proved here; elementary, possibly folklore)**.** *For $k \ge 2$ and $0 \le \varepsilon < 1$,*

$$M_{k,\varepsilon} \le (1+\varepsilon)M_k \le (1+\varepsilon)\frac{k}{k-1}\log k.$$

Conversely, for $0 \le \varepsilon \le M_k/(4k)$,

$$M_{k,\varepsilon} \ge (1+\varepsilon)M_k - \frac{64k^2}{M_k}\,\varepsilon^2.$$

In particular $\lim_{\varepsilon\to 0^+}(M_{k,\varepsilon} - M_k)/\varepsilon = M_k$.

Proof. By monotonicity in τ and scaling, $M_k(1+\varepsilon, 1-\varepsilon) \le M_k(1+\varepsilon, 1+\varepsilon) = (1+\varepsilon)M_k$, and $M_k \le \frac{k}{k-1}\log k$ is [14, Cor. 6.4].

For the lower bound, scaling gives $M_{k,\varepsilon} = (1+\varepsilon)M_k(1,\rho)$ with $\rho = (1-\varepsilon)/(1+\varepsilon)$. Put $\delta = k(1-\rho)$ and let F on $\mathcal{R}_k$ satisfy $\sum_m J_k^{(m)}(F) \ge (M_k - \eta)I_k(F)$. Split $F = F_1 + F_2$ with $F_2 = F\mathbf{1}_{S>\rho}$, and let $x = I_k(F_2)/I_k(F)$. Two facts follow from the Cauchy–Schwarz inequality on segments of length at most $1-\rho$. First, a fiber of F_2 has length at most $1-\rho$, so $\sum_m J_k^{(m)}(F_2) \le \delta\, I_k(F_2)$. Second, a fiber over t' with $|t'| > \rho$ lies in $\{S > \rho\}$ and has length at most $1-\rho$, so restricting the marginals of F to $\rho\mathcal{R}_{k-1}$ loses at most $\delta\, I_k(F_2)$. By the triangle inequality for $F \mapsto (\int F\,dt_m)_{m\le k}$ and the first fact,

$$M_k - \eta \le \left(\sqrt{M_k(1-x)} + \sqrt{\delta x}\right)^2 \le M_k(1-x) + 2\sqrt{M_k\delta x} + \delta x.$$

This is a quadratic inequality in $\sqrt{x}$. If $\delta \le M_k/2$ it gives $x \le (16\delta + 8\eta)/M_k$. By the second fact the quotient of F for $M_k(1,\rho)$ is at least $M_k - \eta - \delta x$. Letting $\eta \to 0$ we get $M_k(1,\rho) \ge M_k - 16\delta^2/M_k$. Finally $\delta = 2k\varepsilon/(1+\varepsilon)$, and $\delta \le M_k/2$ when $\varepsilon \le M_k/(4k)$. □

So for fixed k the factor $1+\varepsilon$ is exact to first order, and no dual argument can improve it there. Any saving is an effect of large k: it needs ε of size about $1/k$ or more, where the shell $\rho < \sigma \le 1$ carries most of the mass. The first inequality already restricts the parameter: with Bombieri–Vinogradov the criterion $M_{k,\varepsilon} > 4$ forces $\varepsilon > \varepsilon_0(k) = \frac{4(k-1)}{k\log k} - 1$ (Computed):

k	49	48	45	40	39	35	30
$\varepsilon_0(k)$	0.0068	0.0117	0.0274	0.0572	0.0638	0.0929	0.1369

For larger ε we use the lemma with a weight that changes shape where the indices start to become inactive.

Theorem 15 (Proved here for the inequality; Computed, interval arithmetic, for the numerical statement)**.** *Let $k \ge 2$, $0 < \tau < s_1 \le L$ and $a \ge s_1$. Then*

$$M_k(L,\tau) \le \sup_{0\le\sigma\le\tau} W(\sigma), \qquad W(\sigma) = \frac{ka}{k-1}\log\left(1 + \frac{(k-1)(s_1-\sigma)}{a-\sigma}\right) + L - s_1 + \sigma\log\frac{L-\sigma}{s_1-\sigma}.$$

Consequently $M_{k,\varepsilon} < 3.99$ for every $k \le 29$ and every $0 \le \varepsilon < 1$. In particular the criterion of [14, Thm. 3.12] *on the ε-enlarged simplex cannot give* DHL$[k,2]$ *for any $k \le 29$ from the Bombieri–Vinogradov theorem, nor from any level of distribution $\theta \le 0.501$, whatever the value of ε.*

Proof. Fix $\Lambda > 0$, put $\lambda = \Lambda/(ka)$ and take the two-zone weight

$$\psi_m(t) = \lambda\,(a - S + kt_m) \quad (S \le s_1), \qquad \psi_m(t) = \Lambda\,\frac{t_m}{S} \quad (S > s_1).$$

It is positive wherever $t_m > 0$. Where $S \le s_1 \le a$ every term is non-negative, so the sum over the active indices is at most the sum over all indices, which is $\lambda(k(a-S)+kS) = \Lambda$. Where $S > s_1$ the sum over the active indices is at most $\Lambda \sum_m t_m/S = \Lambda$. On the fiber over t' with $|t'| = \sigma \le \tau$ we have $S = \sigma + u$, and

$$\int_0^{s_1-\sigma} \frac{du}{\lambda(a - \sigma + (k-1)u)} + \int_{s_1-\sigma}^{L-\sigma} \frac{(\sigma+u)\,du}{\Lambda u} = \frac{W(\sigma)}{\Lambda}.$$

Lemma 13 with $\Lambda = \sup W$ gives the inequality.

For the numerical statement it suffices to treat $k = 29$. For $\varepsilon \le 143/1000$, Proposition 14 gives $M_{29,\varepsilon} \le 1.143 \cdot \frac{29}{28} \log 29 < 3.9863$. The interval $[143/1000, 1]$ is cut into 857 intervals $[e, e']$ of length $1/1000$. For each one the file `code/cert_floor_k29.json` lists rational a and s_1 with $\sup W < 3.99$ for $(L, \tau) = (1 + e', 1 - e)$; by monotonicity in L and τ this covers every $\varepsilon \in [e, e']$. The supremum over σ is bounded rigorously: the first logarithm is non-increasing in σ (because $a \ge s_1$) and the last term is non-decreasing, so on a piece $[x, y]$ we bound W by the first term at x plus the last at y, in ball arithmetic at rational points, bisecting until the bound is below 3.99. The script is `code/dual_eps_cert.py`; a second script, `code/check_floor_cert.py`, re-checks the certificate with a different interval library. The largest value of $\sup W$ over the 857 intervals is 3.9772 (floating point). □

Remark 16 (what was known, and a caution). Polymath8b state the uniform bound $M_{k,\varepsilon} \le \frac{k}{k-1}\log(2k-1)$ for $0 \le \varepsilon < 1$ [14, Prop. 6.5], and remark that the bounds for $M_{k,\varepsilon}$ "do not appear to be as sharp". That bound is below 4 exactly for $k \le 23$, so Theorem 15 moves this floor from 24 to 30. It does not use their bound. We note, with caution, that we could not complete the last step of the proof of [14, Prop. 6.5] as we read it: the weights $1 - S + kt_m$ are summed, and for $S > 1$ their sum over the active indices can exceed k (at $t = (s, 0, \dots, 0)$ with $1 < s \le 1 + \varepsilon$ only the first index is active and its weight is $1 + (k-1)s$). Our reconstruction gives the constant $\frac{k}{k-1} + \varepsilon$ in place of $\frac{k}{k-1}$. For version 0.4 we re-read the proof in the current arXiv text: after the Cauchy–Schwarz step it says only "Integrating in $t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k$ and summing in i, we obtain the claim." The weights $1 - S + kt_i$ of the inactive indices can be negative when $S > 1$, and dropping them is what lets the sum exceed k; it is at most $k + (k-1)(S-1) \le k + (k-1)\varepsilon$, which gives our constant. So the argument as printed proves $M_{k,\varepsilon} \le \left(\frac{k}{k-1} + \varepsilon\right)\log(2k-1)$. Whether the stated inequality is nevertheless true is Open; we have not seen the journal version, and we may be missing an intended step. A brief search (the paper [14] and the Polymath wiki page on the variational problem) found no sharper published upper bound for $M_{k,\varepsilon}$; the search was not exhaustive, and Theorem 15 has not been reviewed by anyone outside this project.

Remark 17 (Computed, floating point only: how far this weight goes). The best value of $\sup W$ over (a, s_1), maximised over ε, is about 3.975 for $k = 29$, 4.004 for $k = 30$, 4.257 for $k = 40$ and 4.459 for $k = 50$, in each case near $\varepsilon = 0.24$ to 0.28; the bound of [14, Prop. 6.5] is 4.187, 4.218, 4.482 and 4.689. For small ε the two-zone weight does not beat the scaling bound: at $k = 50$ and $\varepsilon = 1/25$ both give 4.15 (4.1458 against 4.1515), while Polymath8b's certified lower bound is 4.0043. The true threshold is presumably close to $k = 50$, so the theorem is far from sharp. Script: `code/dual_eps.py remark15`.

Remark 18 (a weight that did not work). We also tried the weight $\psi_m = \frac{\beta}{n}(a - S_A + nt_m)$, where n is the number of active indices at t and S_A the sum of the active coordinates. Its sum over the

active indices is exactly βa everywhere, so the bound is a times the largest fiber integral. Evaluating the fiber integral exactly on the fibers over $t' = (x, \dots, x, 0, \dots, 0)$ (COMPUTED, floating point, $k = 50$, $\varepsilon = 1/25$, every $a \in [1.05, 3]$ on a grid of step 0.01, script `code/dual_eps.py remark16`) shows that this weight cannot give less than 4.178, which is worse than the 4.1515 from scaling. The binding fiber has a single non-zero coordinate $x \approx 2\varepsilon$: there the zero coordinates become inactive while n is still large, and the normalisation by n wastes the slack. A dual bound that is sharper for small ε must treat the small coordinates near such sparse points differently. This is OPEN. (An earlier draft of this remark, version 0.3 as first circulated, reported a weaker obstruction, 4.146, from fibers with about ten non-zero coordinates; the exact fiber evaluation above supersedes it.)

The limit of weights that see only t_m and S (version 0.4). Every weight used so far has the form $\psi_m(t) = \Psi(S - t_m, t_m)$: Polymath8b's $\lambda(a - S + kt_m)$ and both zones of Theorem 15. Call this family $\mathcal{W}$. For weights in $\mathcal{W}$ the problem "minimise the bound of Lemma 13" is convex: each fiber condition $\int du/\Psi \le 1$ is convex in Ψ, and each point t gives a linear condition $\sum_{m \text{ active}} \Psi(S - t_m, t_m) \le \Lambda$. So its optimum can be computed, and it tells us how far any weight in $\mathcal{W}$, however clever, can go.

Proposition 19 (COMPUTED, floating point, discretised; not a rigorous bound)**.** *Discretise $\Psi = \rho \cdot (L - \sigma + (k-1)u)$ with ρ constant on cells of an $N \times$ (geometric) grid in (S, u), impose the fiber conditions exactly on the grid $\sigma \in \frac{L}{N}\mathbb{Z}$, and impose the pointwise condition on all configurations of the grid of step L/N (found by a knapsack oracle). The convex optimum $B_N(k, \varepsilon)$, computed by cutting planes with an exact convex solver at each step, is:*

(k, ε)	$N = 80$	$N = 120$	$N = 160$	*best weight known*	$M_{k,\varepsilon} \ge$
$(50, 0)$	*3.9919*			*3.9919 (Cor. 6.4)*	*3.9274*
$(50, 1/25)$	*4.1427*	*4.1420*	*4.1418*	*4.1458 (Thm. 15)*	*4.0012*
$(30, 1/4)$	*4.0501*	*4.0345*	*4.0240*	*4.0021 (Thm. 15)*	–

At $\varepsilon = 0$ the optimum over the whole family is the classical weight: the solver, started anywhere, returns $\frac{k}{k-1}\log k$. At $(50, 1/25)$ the values have settled near 4.14: no weight that sees only t_m and S can bring the dual bound much below 4.14 there, while the true value is at least 4.0012. At $(30, 1/4)$ the discretisation has not converged (the values still fall, and lie above the continuum value ≤ 4.0021 that Theorem 15 already certifies), so we draw no conclusion about the exact floor of $\mathcal{W}$; runs at $k = 31, 32$ gave $B_{120} = 4.0261$ to 4.0976, also unconverged. The lesson is the first one: *to approach the truth near $k = 50$, a dual weight must depend on more of t than t_m and S.* The optimal weight is known in principle: if $F > 0$ is the extremal function, $\psi_m = \int F\, dt_m / F$ makes Lemma 13 an equality (the Collatz–Wielandt principle for the positive operator $F \mapsto \sum_{m \text{ active}} \int F\, dt_m$). A first test (COMPUTED, floating point, 20000 random points, not a bound; `code/cw_sample.py`) is encouraging on the plain simplex and discouraging on the enlarged one. For $k = 3$, $\varepsilon = 0$ and the degree-12 Galerkin function ($M_3 \ge 1.646440$) the sampled ratio stays in $[1.6464, 1.6465]$, which would be an almost sharp upper bound if the supremum could be certified. For $\varepsilon > 0$ the Galerkin polynomial changes sign ($\min F / \max F = -0.0003$ at $(k, \varepsilon, d) = (10, 1/5, 12)$, -0.11 at $(50, 1/25, 12)$). This is structural: where no index is active the extremal function vanishes, so near that region any positive F makes the ratio large. Making the principle usable there is OPEN. The script is `code/dual_lp.py`; `runs/dual_lp_conv.log` and `runs/dual_lp_floor.log` hold the output. The last column is from Table 2 and Theorem 6.

Remark 20 (heuristic: does the enlarged simplex stop at $k = 50$?). The certified values of Table 2 grow with the degree roughly geometrically. Fitting $M(D) = M_\infty - Cr^D$ to three consecutive even degrees (Aitken's Δ^2, script `code/extrapolate_eps.py`) can be tested where

the answer is known. From the COMPUTED values at $D = 16, 18, 20$ for $k = 50$, $\varepsilon = 1/25$ it predicts 3.99324, 4.00195, 4.00553 at $D = 22, 25, 27$, against the certified 3.99310 (degree 22), our 4.00125 (degree 25) and Polymath8b's 4.0043 (degree 27). The model is slightly optimistic, by 0.0001 to 0.0012. Applied to $k = 49$ with $D = 18, 20, 22$ (certified 3.97715 and 3.97690 at degree 22 for $\varepsilon = 1/30$ and $1/25$) it gives limits 3.9922 and 3.9938. As a test registered before the run, the fit predicted 3.98230 at degree 24 for $\varepsilon = 1/30$; the certified value is 3.98218 (1508 coefficients), again slightly below, and the fit through $D = 20, 22, 24$ moves the limit down to 3.9913. The grid $\varepsilon \in \{1/50, 1/40, 1/30, 1/25, 1/20\}$ at $D \le 20$ gives at most 3.995. This suggests that $\sup_\varepsilon M_{49,\varepsilon} < 4$, so that the ε-enlarged simplex with Bombieri–Vinogradov stops at $k = 50$, consistent with the fact that every bound below 246 uses a different support. This is a heuristic, not a theorem: OPEN. Together with Theorem 15 it brackets the floor of this method between 30 and (conjecturally) 50.

Remark 21 (scope). Theorem 15 concerns the ε-enlarged simplex only. It says nothing about the 2026 claims, which use other supports and new equidistribution estimates. A floor for those is the Problem in the next section.

5. PROGRAMME

Track A: lowering the bound.

(1) *Climb the ladder.* The certified pipeline now stands at $k = 50$ ($H_1 \le 246$, the established bound, via the ε-enlarged simplex). Next: Stadlmann's region ($k = 49$, 240), which needs the rough-mass caps of [18] in the integration formulas.

(2) *Floor of the method (our main original direction).* For the plain simplex the dual bound $M_k \le \frac{k}{k-1} \log k$ shows where the method stops ($k \ge 51$). For the ε-enlarged simplex Theorem 15 gives $k \ge 30$, which is still far from the expected truth near 50; Section 4 explains what a sharper weight must do, and Proposition 19 indicates that it must depend on more of t than t_m and S (a Collatz–Wielandt weight $\psi_m = \int F\, dt_m / F$ built from a good positive test function F is the natural candidate). No upper bound of this kind is known to us for the 2026 hybrid supports. A rigorous dual certificate there would say whether 182 is close to the limit of the present inputs or far from it, and which input (which Type I/II/III range) is binding.

(3) *Which rung is cheap.* From Table 1, the gain per unit of k below the claimed frontier is $182 \to 176 \to 168 \to 162 \to 158$ for $k = 39, \dots, 35$: the steps $39 \to 38$ and $38 \to 37$ are worth the most.

Problem 22 (OPEN)**.** Give a rigorous upper bound for the variational quantity of [18] as a function of k and of the admissible exponents, and deduce the least k the hybrid method can reach with the currently proved estimates.

Track B: the parity barrier. No sieve of the above kind can distinguish integers with an even number of prime factors from those with an odd number. Known ways around the barrier all inject bilinear or structural information. We will study, one per session, what each would need in order to say something about twin primes: Chen's theorem [4]; Bombieri's asymptotic sieve [2]; the parity-breaking sieve of Friedlander and Iwaniec [6]; Heath-Brown's theorem that Siegel zeros would imply the conjecture [7]; the function-field proof of Sawin and Shusterman [16]; and the entropy-based results on correlations of multiplicative functions [19].

Problem 23 (OPEN, to be made precise in Lean)**.** State, as a single explicit hypothesis on bilinear sums over $n(n+2)$, the weakest known input that implies the twin prime conjecture, and formalise the implication.

Rules of this paper. A numerical bound enters the "established by this project" column only with an exact certificate and, for the combinatorial part, a Lean proof. Ideas are logged with their outcome, including failures. Nothing here is peer reviewed.

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